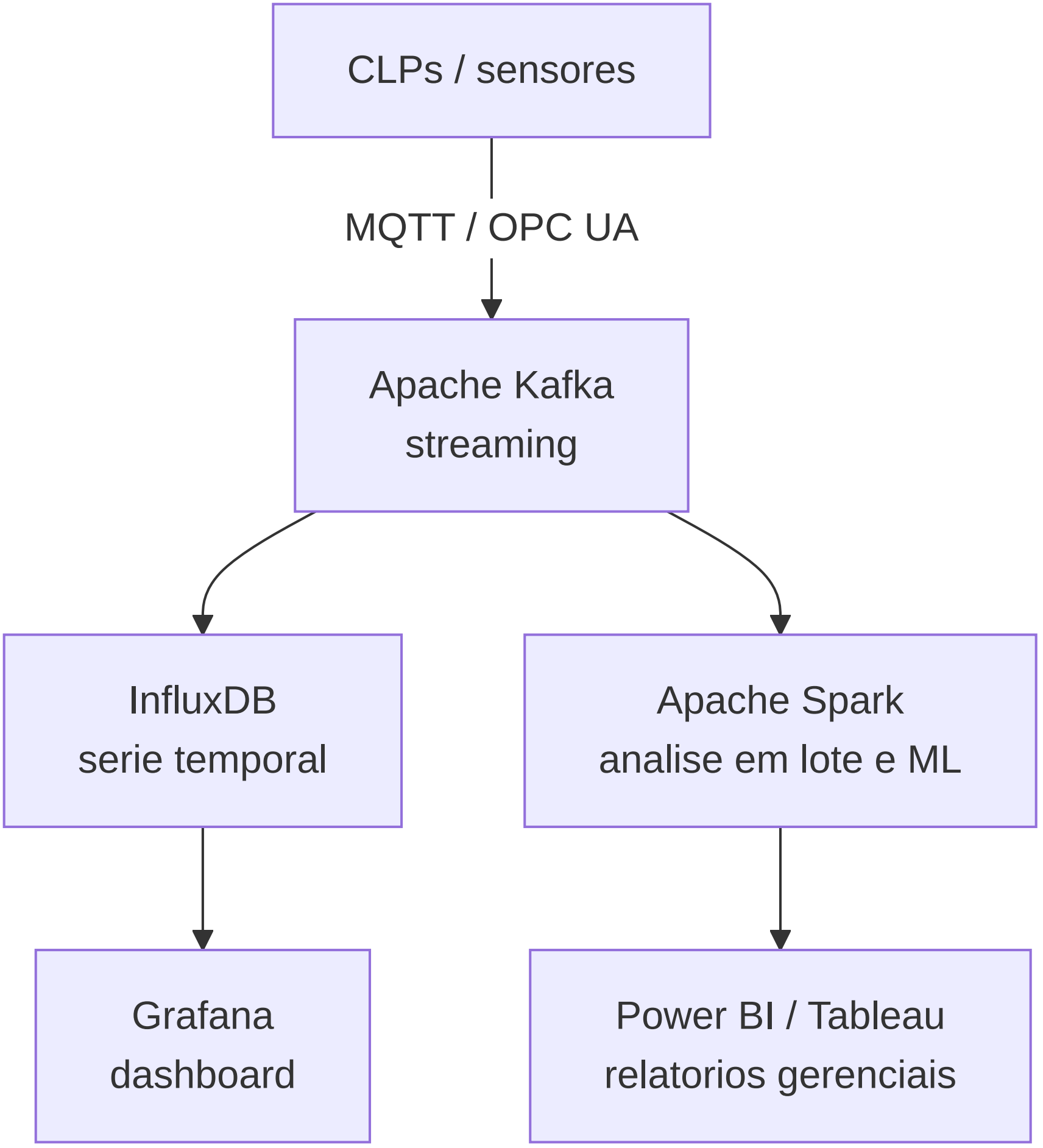
CLPs / sensores
MQTT / OPC UA
Apache Kafka
streaming
InfluxDB
serie temporal
Apache Spark
analise em lote e ML
Grafana
dashboard
Power BI / Tableau
relatorios gerenciais